

Matthew House

AI SOLUTIONS ENGINEER & CONSULTANT

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AI Solutions Engineer building production agentic AI systems: LLM applications, RAG pipelines, and data enrichment. Works end-to-end from scoping through deployment, bridging technical and business to ship AI that drives measurable operational improvement.

TECHNOLOGIES

Languages

Python TypeScript SQL Bash

AI / LLM

LangGraph RAG MCP OpenAI Claude Pydantic AI LiteLLM Langfuse

Application

FastAPI React Next.js Chainlit Streamlit

Data

PostgreSQL SQLAlchemy Vector DBs

Infrastructure

AWS Azure Terraform Docker GitHub Actions N8N

Dev Tooling

Claude Code Cursor Playwright

EDUCATION

 **MEng Aerospace Engineering**
University of Southampton · 2015–2019 · 2:1

CERTIFICATION

 **Azure Data Science Associate**
Microsoft

APPROACH



Ownership

Take projects from initial requirements through to production, working independently or in small teams.



Collaborative

Work directly with founders and teams to understand problems and turn business needs into production solutions.



Communicator

Explain complex technical topics clearly to different audiences, whether executives, engineers, or clients.

Principal AI Solutions Engineer

Jul 25 – Present

Full-time

Lucanet

Lead AI solutions across GTM and Operations, building LLM applications, RAG systems, and automation pipelines. Drive buy vs build analysis and vendor evaluation.

- Built **IT Service Desk AI Agent** in 3 months, multi agent system handling triage, troubleshooting, and knowledge retrieval, **achieving 40% ticket deflection**
LangGraph, Pydantic, Chainlit, Anthropic, PostgreSQL, AWS ECS, Terraform
- Architected **RAG infrastructure** syncing Confluence to git then OpenAI vector stores via GitHub Actions, with metadata and retrieval methods powering assistants across IT, Customer Success, and HR
- Built **GTM data enrichment and territory scoring pipelines** integrating Clay, Gong, Chilli Piper, and Salesforce, improving lead prioritisation
- Drive **AI enablement** through workshops on n8n automation, Copilot Studio, and prompt engineering with playbooks for safe scaling

AI Engineer

Jan 25 – Jun 25

Contract

Connecting Dots

AI and data solutions for clients across ESG, sales intelligence, and operations.

- Built **LLM platform** helping ESG analysts extract climate strategy insights from corporate disclosures, accelerating investment research, discovery to MVP in **2 months**
LangChain, FAISS, Streamlit, PostgreSQL, AWS ECS
- Accelerated document processing via **parallelised inference pipelines, achieving 10x speedup** for investment decisions
- Built **AI outbound pipeline** scraping web and LinkedIn via Apollo, enriching leads with NLP, then generating tailored cold emails

Data Scientist

Nov 20 – Dec 24

Full-time

Atkins Réalis

Data science and automation solutions for infrastructure, environmental, and defence clients.

- Built ARROW, AI optimisation tool using **evolutionary algorithms (NSGA-II)** for wastewater planning for a large UK water provider, won **Atkins 2023 Innovation Award**, Highly Commended **UK IT Industry Awards Data Science Project of the Year**
Python, Streamlit, Azure, ICM InfoWorks
- Automated **wildlife monitoring** using computer vision (YOLOv5, MegaDetector), **reducing manual review time by 80%** for environmental assessments
- Built **predictive maintenance models** using historical failure data and Monte Carlo simulations, reducing unplanned downtime for defence clients